Module 5 Homework

1i. Yes. Heteroskedasticity makes the usual OLS standard errors incorrect, so the resulting t-statistics do not follow the t distribution under H_0.

1ii. No. While a high correlation between explanatory can increase standard errors it does not invalidate the usual OLS t-statistics if there is no perfect collinearity.

1iii. Yes. Omitting an important explanatory variable causes bias in the OLS estimators, so the t-statistics are no longer valid for inference.

2i. The null hypothesis is, holding *sales* and *roe* fixed, *ros* has no effect on CEO salary, (H_0: $\beta_3 = 0$). The alternative hypothesis is, holding *sales* and *roe* fixed, increases in *ros* increases CEO salary (H_1: $\beta_3 > 0$).

2ii. A 50 point increase in *ros* is predicted to increase *salary* by .00024*50*100 = 1.2%. A 50 point increase in *ros* is very large relative to a practically low increase in *salary*.

2iii. Due to the large n = 209 value, the critical value for the 10% significance value is approximately 1.28. The test statistic, (.00024/.00054 = .44) is less than the critical value. Therefore, we fail to reject the null hypothesis. At the 10% significance level, there is insufficient evidence that *ros* has a positive effect on CEO salary.

2iv. Though the coefficient on *ros* is small and the effect is insignificant at the 10% level, it may be correlated with the other included variables, specifically *sales.* As such, there is an argument to be made that including it as a control could avoid bias in the other variables and strengthen the model overall.

5i. The 95% confidence interval for β_{hsGPA} is $0.412 \pm (1.96 * .094)$ = (0.228, 0.596)

5ii. No. The value 0.4 lies within the 95% CI from part 5i. Therefore, we fail to reject the null hypothesis at the 5% level.

5iii. Yes. The value 1.0 lies outside of the 95% CI from part 5i. Therefore, we can reject the null hypothesis at the 5% level.

C1i. Holding *expendB* and *prtystrA* fixed, a 1% increase in Candidate A's campaign expenditures is predicted to change *voteA* by $\beta_1/100$ percentage points.

C1ii. A null hypothesis that a 1% increase in A's expenditures is offset by a 1% increase in B's expenditures is represented as $H_0: \beta_1 + \beta_2 = 0$.

C1iii.

$$\widehat{voteA} = \underset{(3.926)}{\mathbf{45.079}} + \underset{(0.382)}{\mathbf{6.083}}(lexpendA) + \underset{(0.379)}{\mathbf{-6.615}}(lexpendB) + \underset{(0.062)}{\mathbf{0.152}}(prtystrA)$$

$$N = 173 \qquad R^2 = 0.793$$

Yes, both *lexpendB* and *lexpendA* have p values <.0001 indicating statistically significant evidence that they affect the outcome of *voteA*. However, this model cannot be utilized to test the hypothesis in C1ii because only the individual coefficients are tested.

C1iv. The test produces an F-statistic of 1.00 with a p-value of approximately .320. This means that we fail to reject the null hypothesis that a 1% increase in A's expenditures is offset by a 1% increase in B's expenditures.

C9i.

$$\log\widehat{(psoda)} = \underset{(0.294)}{\mathbf{-1.463}} + \underset{(0.031)}{\mathbf{0.073}}(prpblck) + \underset{(0.027)}{\mathbf{0.137}}(\log(income)) + \underset{(0.133)}{\mathbf{0.380}}(prppov)$$

$$N = 401 \qquad R^2 = 0.087$$

β_1 shows a p-value of 0.018 indicating that it is statistically different from zero at the 5% level but not at the 1% level.

C9ii.

The correlation between *lincome* and *prppov* is −0.838, indicating a strong negative relationship between income and poverty rates. In the regression from part C9i, both variables are statistically significant. The two-sided p-value for *lincome* is <0.0001, and the p-value for *prppov* is 0.0044.

C9iii.

$$\widehat{\log(psoda)} = \underset{(0.292)}{\mathbf{-0.842}} + \underset{(0.029)}{\mathbf{0.098}}(prpblck) + \underset{(0.038)}{\mathbf{-0.053}}(\log(income)) + \underset{(0.135)}{\mathbf{0.052}}(prppov) + \underset{(0.018)}{\mathbf{0.121}}(lhseval)$$

$$N = 401 \qquad R^2 = 0.184$$

Holding *prpblck*, *income*, and *prppov* constant, a 1% increase in *lhseval* is predicted to increase *psoda* by 0.12%. The two-sided p-value for *lhseval* is < 0.0001, so we reject the null hypothesis that the coefficient on *lhseval* is zero.

C9iv. When compared to part C9i, the coefficient on *prpblck* increases from 0.073 to 0.098 and its p-value falls from 0.018 to 0.0009. The coefficient on log(*income*) decreases from 0.137 to −0.053, and its p-value rises from <0.0001 to 0.1587. Testing the null hypothesis of joint significance between log(*income*) and *prppov* produces a p-value of 0.0304. At the 5% level, we reject the null hypothesis. On their own, *lincome* and *prppov* are insignificant once *lhseval* is included, the test shows they are jointly significant.

C9v. The 2nd regression is more reliable in determining whether the racial makeup of a zip code influences local fast-food prices. Including the significant variable *lhseval* reduces bias in the other coefficients. Furthermore, the 2nd regression sees an increase in the R^2 from .087 to .184.

C11i.

$$\widehat{\mathrm{educ}} = \underset{(0.287)}{\mathbf{8.240}} + \underset{(0.028)}{\mathbf{0.190}}(motheduc) + \underset{(0.020)}{0.\mathbf{109}}(fatheduc) + \underset{(0.030)}{\mathbf{0.402}}(abil) + \underset{(0.008)}{\mathbf{0.051}}(abil2)$$

$$N = 1230 \qquad R^2 = 0.444$$

Testing the null hypothesis that *educ* is linearly related to *abil* ($\beta_4 abil^2 = 0$) yielded p<0.0001 so there is significant evidence to reject the null hypothesis and that a quadratic relationship exists between *abil* and *educ*.

C11ii. Testing the null hypothesis that $\beta_1 = \beta_2$ yielded a p-value of 0.0531. At the 5% significance level, we fail to reject the null hypothesis.

C11iii.

Variable	**Coefficient**	**Std. Error**
Intercept	8.082	0.313
motheduc	0.193	0.028
fatheduc	0.108	0.020
abil	0.399	0.030
$abil^2$	0.051	0.008
tuit17	0.016	0.063
tuit18	0.000	0.064

$$N = 1230 \qquad R^2 = 0.445$$

Testing the null hypothesis that the two college tuition variables are jointly statistically significant yielded a p-value of 0.4322. At the 5% significance level, there is insufficient evidence to reject the null hypothesis.

C11iv. The two variables, *tuit17* and *tuit18* are highly correlated with a value of 0.981. Switching to including these variables as an average of both years removes this issue of multicollinearity. The average tuition is more significant than each year individually. However, with a p-value of 0.197, the evidence still does not suggest that *avgtuit* has a statistically significant impact on *educ*.

C11v. Intuitively, I might have expected tuition to have more of an impact on educational attainment. However, introducing the variables into the model from C11i (where they weren't yet included) resulted in a very minor increase in the R^2 value (from .444 to .445). Since the other variables in the model all show p-values less than .0001, it is possible that much of the variation in *educ* is already captured by family education and ability.